



Technical Service Bulletin

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Fault Code Severity and Fault Lamp Changes

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Technical Service Bulletin announces changes in the:

- Severity of several Quantum™ system fault codes
- Mapping of fault codes to their respective alarm lamp colors on QSK45, QSK60, and QSK78 powered industrial applications.

These changes optimize the Quantum™ high horsepower annunciation and prevent operator confusion over the lamp severity. The lamp color Red now means, STOP the engine immediately, continued operation will cause engine damage.

The improvement affects engine families QSK45, QSK60, and QSK78 built or recalibrated after August 7, 2003. The following table lists specific Quantum™ ECM code revisions affected and revision level. The revision date for these codes is July 8, 2003.

Note : This is a product improvement, not subject to campaign.

ECM Code	ECM Code Revision	SC Option	DO Option
D60010	31	SC06790	DO06066
D60011	29	SC06748	DO06067
D60012	25	SC06796	DO06073
D60015	24	SC06798	DO06075
D60016	28	SC06799	DO06076
D60017	28	SC06809	DO06087
D60018	24	SC06814	DO06092
D60019	27	SC06800	DO06077
D60021	26	SC06774	DO06068
D60022	26	SC06801	DO06078

ECM Code	ECM Code Revision	SC Option	DO Option
D60023	26	SC06802	DO06079
D60024	24	SC06803	DO06080
D60025	24	SC06804	DO06081
D60026	24	SC06805	DO06083
D60028	24	SC06806	DO06084
D60029	25	SC06807	DO06085
D60031	25	SC06808	DO06086
D60033	26	SC06813	DO06091
D60034	32	SC06817	DO06095
D60035	32	SC06818	DO06096
D60036	30	SC06819	DO06097
D60037	30	SC06820	DO06098
D60038	32	SC06844	DO06103
D60039	32	SC06845	DO06104
D60040	32	SC06850	DO06105
D60041	32	SC06851	DO06106
D60042	24	SC06797	DO06074
D60043	28	SC06810	DO06088
D60044	28	SC06811	DO06089
D60045	24	SC06812	DO06090
D60046	26	SC06815	DO06082
D60047	27	SC06893	DO06168
D60048	27	SC06894	DO06169
D60049	29	SC06895	DO06170
D60050	29	SC06896	DO06171
D60051	31	SC06899	DO06174
D60052	31	SC06900	DO06175
D60053	28	SC06909	DO06202
D60054	28	SC06910	DO06203
D60055	31	SC06913	DO06190
D60056	31	SC06914	DO06191
D60059	22	SC06795	DO06072
D60062	26	SC06831	DO06143
D60063	26	SC06832	DO06144
D60064	26	SC06848	DO06139
D60065	26	SC06849	DO06140

ECM Code	ECM Code Revision	SC Option	DO Option
D60066	26	SC06883	DO06157
D60067	26	SC06884	DO06158
D60068	23	SC06825	DO06135
D60069	23	SC06826	DO06136
D60070	22	SC06827	DO06137
D60071	22	SC06828	DO06138
D60072	22	SC06881	DO06155
D60073	22	SC06882	DO06156
D60074	23	SC06889	DO06163
D60075	23	SC06890	DO06164
D60076	27	SC06868	DO06125
D60077	26	SC06869	DO06126
D60078	24	SC06870	DO06127
D60079	24	SC06871	DO06128
D60080	29	SC06877	DO06151
D60081	29	SC06878	DO06152
D60084	24	SC06925	DO06184
D60085	24	SC06926	DO06185
D60086	29	SC06927	DO06188
D60087	29	SC06928	DO06189
D60088	22	SC06821	DO06099
D60089	22	SC06822	DO06100
D60090	22	SC06879	DO06153
D60091	22	SC06880	DO06154
D60092	23	SC06923	DO06200
D60093	23	SC06924	DO06201
D60094	20	SC06942	DO06225
D60095	20	SC06943	DO06226
D60096	22	SC06864	DO06121
D60097	22	SC06865	DO06122
D60098	23	SC06866	DO06123
D60099	23	SC06867	DO06124
D60100	22	SC06872	DO06129
D60101	22	SC06873	DO06130
D60102	22	SC06905	DO06180
D60103	22	SC06906	DO06181

ECM Code	ECM Code Revision	SC Option	DO Option
D60104	23	SC06911	DO06204
D60105	23	SC06912	DO06205
D60106	22	SC06915	DO06192
D60107	22	SC06916	DO06193
D60108	21	SC06854	DO06109
D60109	21	SC06855	DO06110
D60110	24	SC06860	DO06117
D60111	24	SC06861	DO06118
D60112	23	SC06885	DO06159
D60113	23	SC06886	DO06160
D60114	24	SC06897	DO06172
D60115	24	SC06898	DO06173
D60116	21	SC06858	DO06115
D60117	21	SC06859	DO06116
D60118	22	SC06862	DO06119
D60119	22	SC06863	DO06120
D60120	21	SC06907	DO06182
D60121	21	SC06908	DO06183
D60122	19	SC06833	DO06145
D60123	19	SC06834	DO06146
D60124	19	SC06887	DO06161
D60125	19	SC06888	DO06162
D60126	19	SC06934	DO06217
D60127	19	SC06935	DO06218
D60128	19	SC06936	DO06219
D60129	19	SC06937	DO06220
D60130	18	SC06957	DO06232
D60131	18	SC60013	DO06237
D60132	18	SC60014	DO06238
D60135	11	SC06958	DO06223
D60140	12	SC06938	DO06221
D60141	12	SC06939	DO06222
D60142	12	SC06940	DO06223
D60143	12	SC06941	DO06224
D60146	12	SC60054	DO06263
D60147	12	SC60055	DO06264

ECM Code	ECM Code Revision	SC Option	DO Option
D60148	12	SC60056	DO06265
D60149	12	SC60057	DO06266
D60150	10	SC60043	DO06253
D60151	10	SC60046	DO06255
D60152	10	SC60044	DO06252
D60153	10	SC60045	DO06254
D60154	9	SC60066	DO06271
D60155	9	SC60067	DO06272
D60156	9	SC60068	DO06273
D60157	9	SC60069	DO06274
D60158	8	SC60047	DO06256
D60159	9	SC60048	DO06257
D60160	7	SC60050	DO06259
D60161	7	SC60051	DO06260
D60162	7	SC60052	DO06261
D60163	9	SC60053	DO06262
D60166	8	SC60058	DO06267
D60167	8	SC60059	DO06268
D60168	8	SC60060	DO06269
D60169	8	SC60061	DO06270
D60170	11	SC60829	DO06141
D60171	11	SC60830	DO06142
D60172	11	SC60891	DO06165
D60173	11	SC60892	DO06166
D60174	9	SC60077	DO06280
D60175	9	SC60078	DO06281
D60176	8	SC60079	DO06282
D60177	8	SC60080	DO06283
D60178	8	SC60081	DO06284
D60179	8	SC60082	DO06285
D60184	6	SC60102	DO06292
D60185	7	SC06921	DO06198
D60186	6	SC06922	DO06199
D60188	5	SC06931	DO06206
D60189	4	SC60110	DO06293
D60190	4	SC60111	DO06294

ECM Code	ECM Code Revision	SC Option	DO Option
D60191	4	SC60121	DO06297
D60192	4	SC60122	DO06298
D60195	2	SC60128	DO06307
D60196	2	SC60129	DO06308
D60197	2	SC60124	DO06303
D60198	2	SC60125	DO06304
D60199	2	SC60126	DO06305
D60200	2	SC60127	DO06306
D60201	2	SC60130	DO06309
D60202	2	SC60131	DO06310
D60203	3	SC60136	DO06316
D60204	3	SC60137	DO06317

This is the recommended setup for fault annunciation:

1. Pin 18 of the 31-pin OEM connector
2. Level 1 faults, includes engine protection and critical sensor in-range faults
3. Indicates continued operation will cause engine damage
4. Operator **must** stop the engine as soon as safety permits
5. Required lamp
6. Red means stop.

1. Pin 16 of the 31-pin OEM connector
2. Level 2 faults, includes critical sensor out of range faults
3. Indicates critical sensors are **not** functional
4. Operator **must** schedule engine service and have the fault corrected the same day
5. Required lamp, unless separate equipment is provided for monitor and display of level 2 faults.

1. Pin 17 of the 31-pin OEM connector
2. Level 3 faults, includes engine information and non-critical sensor out of range faults
3. Indicates service is required within the next preventative maintenance interval
4. This lamp is **not** required but recommended, unless a separate equipment is provided for monitor and display of level 3 faults.

Depending on the application, some physical wiring changes need to be made to existing equipment when Quantum™ calibration is updated. For three lamp configurations, a correction of the lamp colors is needed for hardware compliance. Two lamp configurations need a minor wiring change. For more details on the action to be taken at the time of software version update, please consult AEB 15.43 and your application engineer or Distributorship Field Service Engineer (DFSE).

The following table lists the hardware configuration required to implement this fault lamp strategy improvement:

Quantum™ Fault Lamp Hardware Changes				
Quantum™ OEM Pin Connection	Previous Function	Previous Lamp Color	New Lamp Function	New lamp Color
18	Engine Protection	Red, Blue, or White (varies by application)	Stop engine immediately	Red
16	Stop	Red	Warning - engine will run, but must be repaired the same day	Amber
17	Warning	Amber	Maintenance - repair at the next planned maintenance	Blue (if used) or none

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